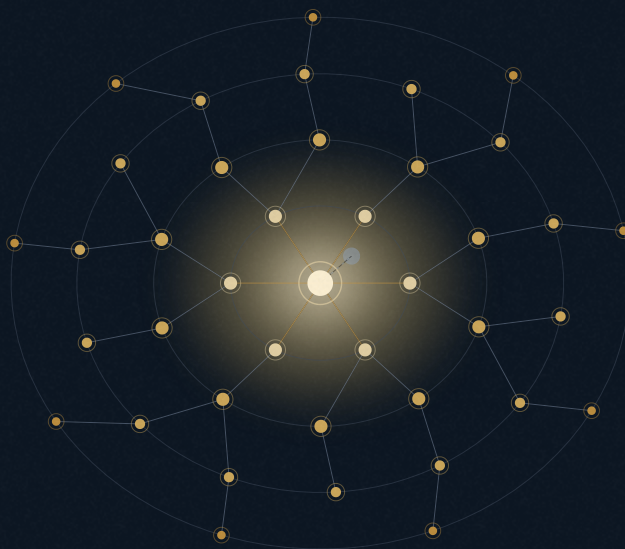




KRONOS FUSION ENERGY



PAPER 2.7 OF THE 2026 KRONOS SET

The Safety Clamp

A Provable Zero-Escape Guarantee

A control-barrier-certified safety filter that wraps any controller with a provable zero-escape guarantee on the operating envelope.

0/5000 escapes clamped vs 5000/5000 · analytic proof

Explore it live – [3-D model](#) · [physics validator](#) · [film series](#)

THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

The Safety Clamp — A Provable Zero-Escape Guarantee

A deterministic, control-barrier-certified safety clamp provides a provable zero-escape guarantee for the compact spherical-tokamak breeder operating envelope. A robust, disturbance-tightened control-barrier-function filter with a closed-form separable solution wraps an arbitrary controller: 0 of 5000 Monte-Carlo trajectories escape the safe set with the clamp versus 5000 of 5000 without it, backed by an analytic forward-invariance proof.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132176](https://doi.org/10.5281/zenodo.22132176) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords:

control barrier function

safety certification

forward invariance

zero-escape

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

Performance inside a proof

A learned or optimization-based controller cannot be allowed to drive a nuclear-regulated machine out of its safe envelope. The clamp guarantees it cannot: a control-barrier filter with a closed-form solution wraps any controller and enforces forward invariance of the safe set.

THE GUARANTEE

0 of 5000 trajectories escape the safe set with the clamp, against 5000 of 5000 without it — backed by an analytic forward-invariance proof, not a training score. Scope is stated openly: the reduced-order model and a declared disturbance bound.

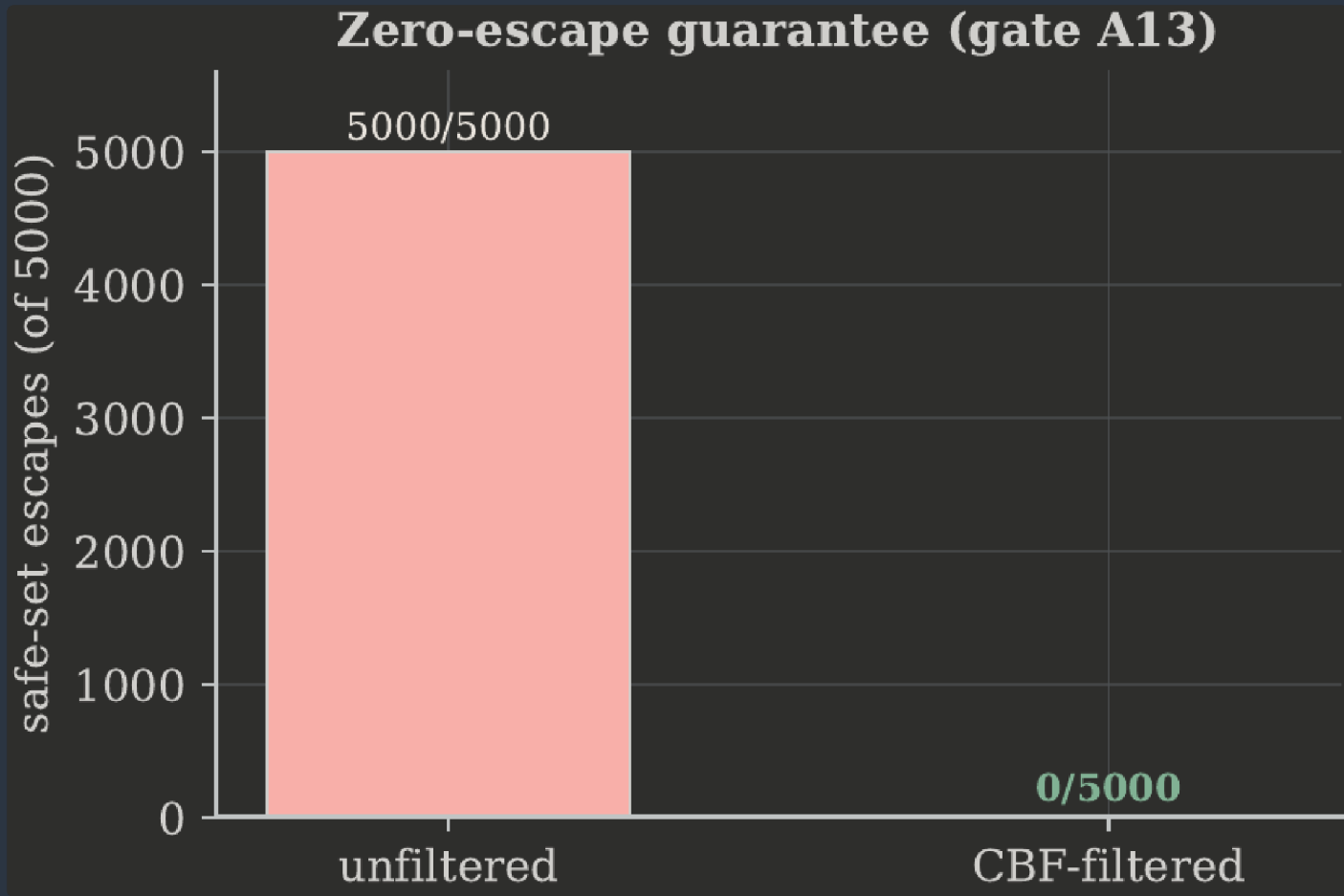


FIGURE 1. Safe-set escapes with and without the clamp over 5000 Monte-Carlo trajectories. MONTE-CARLO

Forward invariance: the clamped trajectory never leaves the safe set

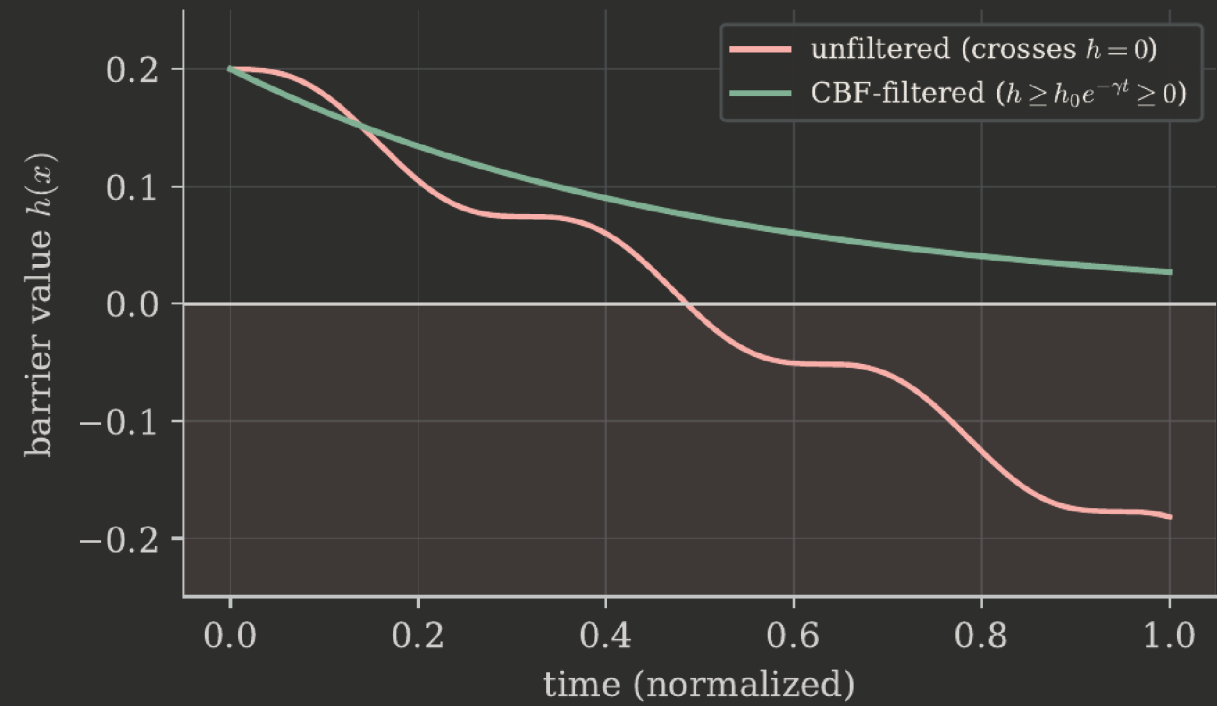


FIGURE 2. The control-barrier filter: forward invariance of the safe set, in closed form. RECOMPUTED

A P P A R A T U S

Portfolio, People & Record

The intellectual-property estate, the people who built it, the limitations that gate it, and the sources it rests on.

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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2022–Present

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REFERENCES

Sources

- [1] H.-S. Bosch and G. M. Hale, “Improved formulas for fusion cross-sections and thermal reactivities,” *Nuclear Fusion* **32**, 611 (1992).
- [2] A. S. Richardson, *NRL Plasma Formulary*, Naval Research Laboratory (2019).
- [3] I. E. Ochs, E. J. Kolmes, and N. J. Fisch, “Bremsstrahlung and the electron–electron contribution in fusion plasmas,” *Physics of Plasmas* (2024).
- [4] T. H. Rider, “Fundamental limitations on plasma fusion systems not in thermodynamic equilibrium,” *Physics of Plasmas* **4**, 1039 (1997).
- [5] ITER Physics Expert Groups, “ITER Physics Basis,” *Nuclear Fusion* **39**, 2137 (1999).
- [6] N. A. Uckan and J. Sheffield, “Tokamak scaling and confinement,” in *Tokamak Concept Improvement*, IAEA (1989).
- [7] O. Sauter, C. Angioni, and Y. R. Lin-Liu, “Neoclassical conductivity and bootstrap current,” *Physics of Plasmas* **6**, 2834 (1999).
- [8] F. Troyon *et al.*, “MHD limits to plasma confinement,” *Plasma Physics and Controlled Fusion* **26**, 209 (1984).
- [9] J. P. Freidberg, *Plasma Physics and Fusion Energy*, Cambridge University Press (2007).
- [10] R. F. Post, “The magnetic mirror approach to fusion,” *Nuclear Fusion* **27**, 1579 (1987).
- [11] D. E. Baldwin, “End-loss processes from mirror machines,” *Reviews of Modern Physics* **49**, 317 (1977).
- [12] T. K. Fowler, R. W. Moir, and T. C. Simonen, “A new simpler tandem mirror,” *Nuclear Fusion* **57**, 056014 (2017).
- [13] G. L. Kulcinski and J. F. Santarius, “Advanced fuels and the D–³He cycle,” University of Wisconsin Fusion Technology Institute.
- [14] W. L. Barr and R. W. Moir, “Test results on direct energy converters for mirror reactors,” *Nuclear Technology/Fusion* **3**, 98 (1983).
- [15] D. B. Go *et al.*, “Thermionic energy conversion in the twenty-first century,” *Frontiers in Mechanical Engineering* **3**, 13 (2017).
- [16] M. Wang *et al.*, “The AME 2020 atomic mass evaluation,” *Chinese Physics C* **45**, 030003 (2021).
- [17] J. E. Menard *et al.*, “Fusion nuclear science facilities and pilot plants based on the spherical tokamak,” *Nuclear Fusion* **56**, 106023 (2016).
- [18] A. J. Creely *et al.*, “Overview of the SPARC tokamak,” *Journal of Plasma Physics* **86**, 865860502 (2020).
- [19] T. P. Wright, “Factors affecting the cost of airplanes,” *Journal of the Aeronautical Sciences* **3**, 122 (1936).
- [20] U.S. Department of Energy / NNSA, *Stockpile Stewardship and Management Plan*, FY2024.
- [21] J. F. Santarius, G. L. Kulcinski, and L. A. El-Guebaly, “A strategy for D–³He fusion energy development,” *Journal of Fusion Energy* **17**, 33 (1998).
- [22] M. A. Abdou *et al.*, APEX advanced blanket concepts, *Fusion Engineering and Design* (1999–2003).
- [23] Federal Register, notice on supply of tritium to non-federal entities (1999).
- [24] Kronos Fusion Energy, “Simulation register and reproducibility deposit,” Zenodo (2026).
- [25] A. O. Thome *et al.*, “ELM-free negative-triangularity edge on DIII-D,” *Plasma Physics and Controlled Fusion* (2024), doi:10.1088/1361-6587/ad6f40.
- [26] T. Eich *et al.*, “Scaling of the tokamak near scrape-off layer power width,” *Nuclear Fusion* **53**, 093031 (2013).
- [27] J. Lim *et al.*, “Scrape-off-layer width in negative-triangularity plasmas,” *Nuclear Fusion / Plasma Physics and Controlled Fusion* (2023).

[28] A. O. Nelson *et al.*, “Prohibition of second-stability access at negative triangularity,” *Plasma Physics and Controlled Fusion* **64**, 124002 (2022).

COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control · 10.5281/zenodo.21842371

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The Safety Clamp

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kronosfusionenergy.com/physics-Validation-Simulation

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